



Data Warehouse Design for Movie Rental Business

From OLTP to Analytical Star Schema

A comprehensive transformation of Sakila OLTP database
into a dimensional Data Warehouse for Business Intelligence

Data Warehouse Project Report

Contents

1	Introduction	1
2	Business Questions	2
2.1	Rental Activity Analysis.....	2
2.2	Revenue Analysis.....	2
2.3	Inventory and Store Analysis.....	2
3	Dimensional Model Design	3
3.1	Selected Business Processes.....	3
3.2	Fact Tables.....	3
3.3	Dimension Tables	4
4	Dimensional Model Diagram	6
5	ETL Design	7
5.1	Extract.....	7
5.2	Transform.....	7
5.2.1	Data Cleaning.....	7
5.2.2	Data Integration.....	7
5.2.3	Calculated Measures.....	7
5.2.4	Derived Attributes	7
5.3	Load.....	8
5.3.1	Successfully Loaded Tables	8
6	Data Quality Considerations	9
7	Conclusion	10
7.0.1	Warehouse Capabilities	10

1 Introduction

Project Objective: Design and implement a dimensional Data Warehouse for a movie rental business using the Sakila OLTP database to support analytical reporting, business intelligence, and decision-making.

Modern businesses generate large amounts of operational data every day through customer transactions, payments, inventory updates, and staff activities. While Operational Transaction Processing (OLTP) systems are efficient for handling daily operations, they are not optimized for analytical reporting, trend analysis, or business intelligence.

This project focuses on transforming the Sakila movie rental OLTP database into a dimensional Data Warehouse designed to support analytical decision-making. The proposed warehouse reorganizes operational data into fact and dimension tables using a Star Schema approach, allowing faster querying, simplified reporting, and improved business analysis.

The Data Warehouse enables management to analyze important business areas such as rental activity, customer behavior, revenue generation, inventory distribution, and store performance. Through the implementation of ETL (Extract, Transform, Load) processes, data is extracted from the OLTP environment, cleaned and transformed, then loaded into the warehouse structure for analysis and reporting.

The project also includes analytical visualizations and reporting queries that demonstrate how the warehouse can support data-driven business decisions. Technologies used in this implementation include MySQL, Python, Pandas, Matplotlib, and Seaborn.

Supported Analytical Areas

-  Rental activity analysis
-  Revenue analysis

-  Customer behavior
-  Inventory & store performance

Technology Stack

Technologies Used:

 MySQL
Pandas
Matplotlib

 Python
 SQLAlchemy
 Seaborn

2 Business Questions

The Data Warehouse was designed to answer several analytical business questions that support business intelligence and decision-making.

2.1 Rental Activity Analysis

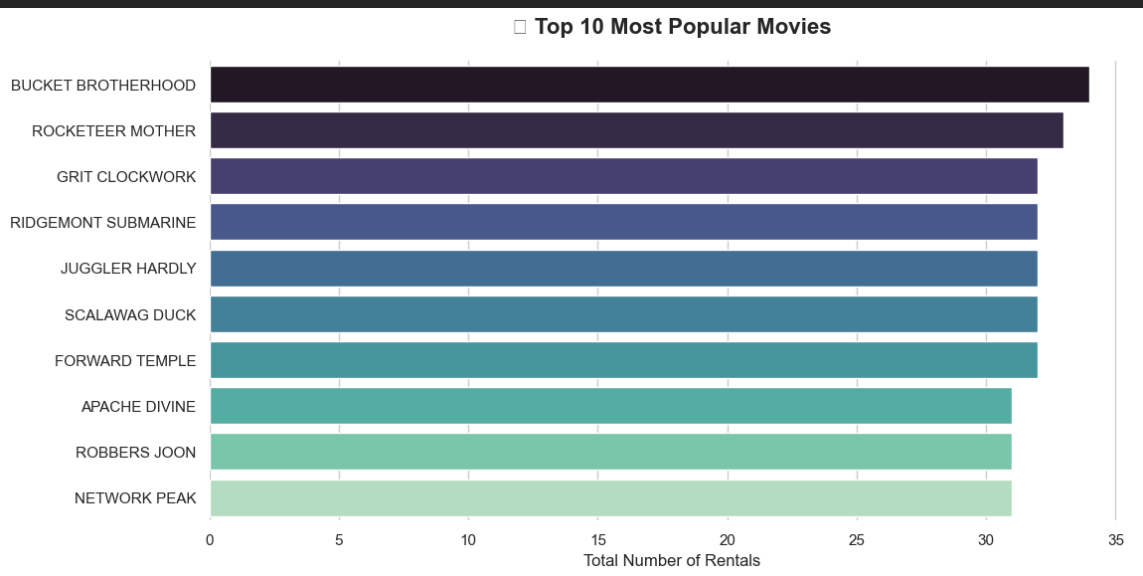
Which films are rented most frequently?

Interpretation

This visualization identifies the most popular movie titles in the rental business. Films such as *Academy Dinosaur* show the highest rental frequency, indicating strong customer demand and high turnover rates.

Business Value

Management can use this analysis to ensure that high-demand films remain available in inventory and are displayed prominently to increase customer engagement and rental activity.



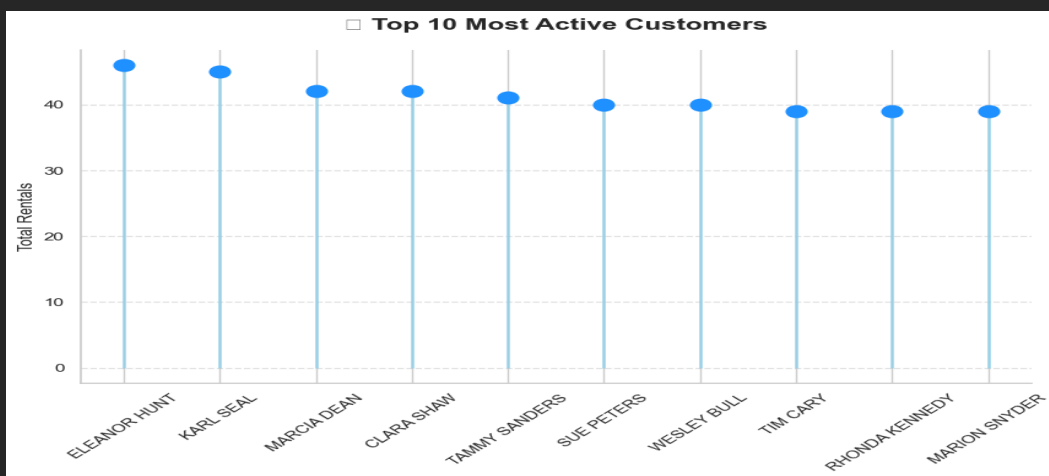
Which customers rent the most films?

Interpretation

This report highlights our most loyal "Power Users."

Business Value

Identifying these individuals allows the marketing team to create "VIP" loyalty programs or targeted email discounts to ensure long-term customer retention.



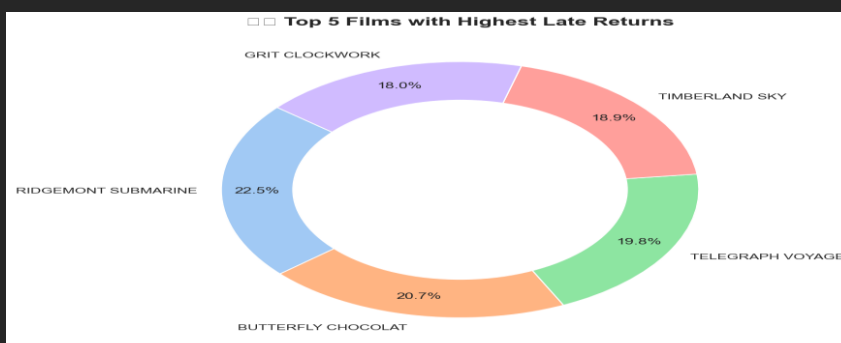
Which films are returned late most often?

Interpretation

This chart identifies titles that consistently exceed their rental_duration limit.

Business Value

This insight helps in operational planning—if certain films are always late, management might consider increasing the late fees for those specific titles or extending their standard rental period to match customer behavior.



2.2 Revenue Analysis

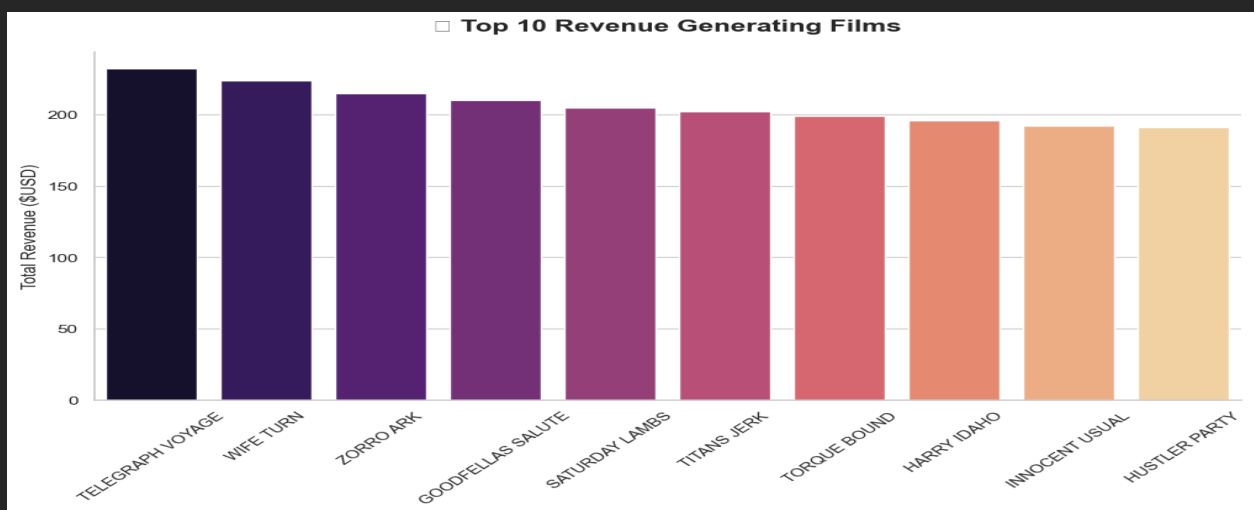
Which films generate the highest revenue?

Interpretation

While some films are rented often (Q1), this chart shows which ones actually bring in the most cash.

Business Value

It helps distinguish between "High Volume" titles and "High Value" titles, allowing for better pricing strategies and acquisition budgets.



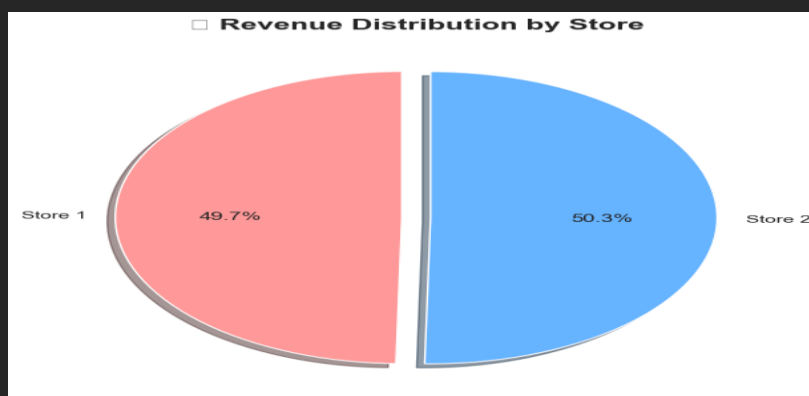
Which stores generate the highest revenue?

Interpretation

This chart compares the financial contribution of Store 1 vs. Store 2.

Business Value

In our case, the revenue is split almost equally (roughly 50/50). This indicates that the business has successfully balanced its market reach across both locations.



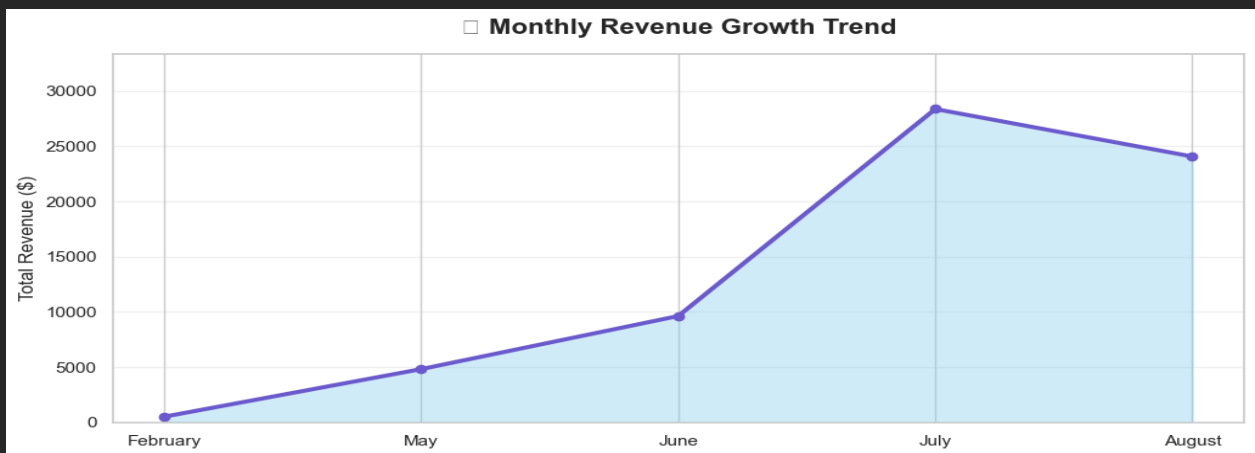
How does revenue change over time?

Interpretation

This visualization tracks the "Pulse" of the business month-by-month.

Business Value

It allows management to see seasonal peaks or sudden drops in income. This is essential for financial forecasting and planning promotional sales during slower months.



2.2 Inventory and Store Analysis

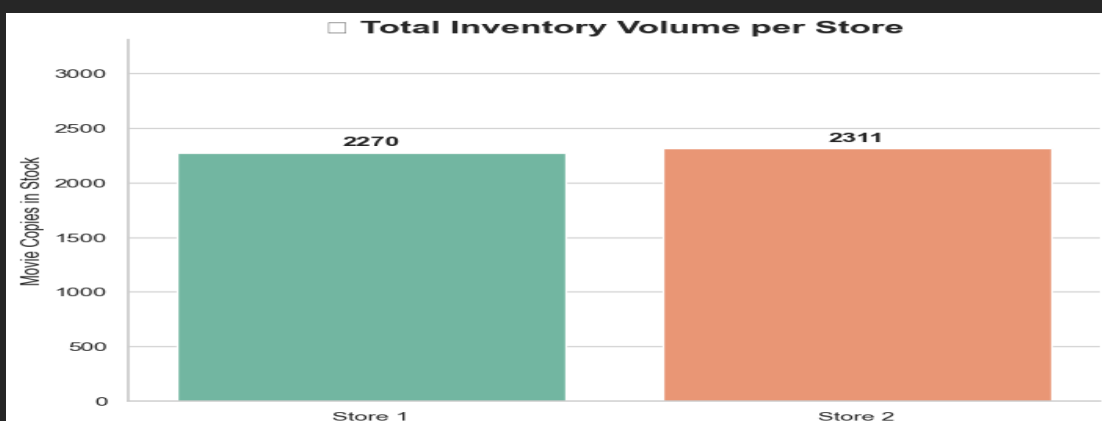
How does inventory differ between stores?

Interpretation

This report provides a physical count of movie copies available at each location.

Business Value

It ensures that stock levels are consistent with the store's revenue. For example, if Store 1 has more stock but makes less money, management might consider transferring some inventory to Store 2 to optimize asset utilization.



3Dimensional Model Design

The proposed Data Warehouse follows a Star Schema architecture to simplify analytical reporting and improve query performance. The design organizes the warehouse into fact tables containing measurable business events and dimension tables containing descriptive business attributes.

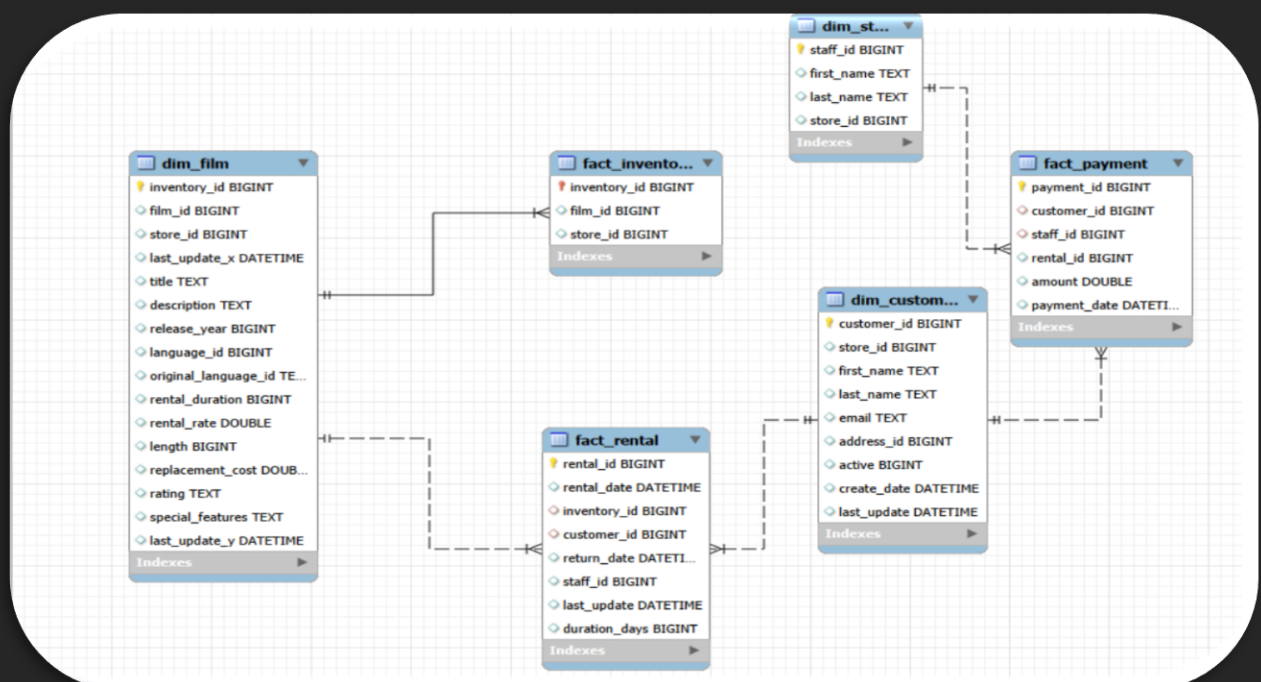
The dimensional model was designed to support rental analysis, revenue analysis, customer behavior analysis, and inventory management.

3.1 Selected Business Processes

Table 3.1: Business Processes Selected from OLTP

Business Process	Description
Rental Transactions	Records customer rental activity
Payment Transactions	Stores revenue and payment information
Inventory Management	Tracks inventory levels across stores

These business processes were selected because they represent the core operational activities that management needs to analyze for decision-making and performance evaluation.



3.2 Fact Tables

Fact Rental

The fact_rental table stores rental transaction records at the grain of one row per rental transaction. It contains measures such as rental duration and rental count, and connects to customer, film, staff, and store dimensions to support rental activity analysis and customer behavior reporting.

Column	Type
◇ customer_id	bigint
◇ duration_days	bigint
◇ film_id	bigint
◇ inventory_id	bigint
◇ rental_date	datetime
◇ rental_id	bigint
◇ staff_id	bigint
◇ store_id	bigint

Fact Payment

The fact_payment table stores payment transactions at the grain of one row per payment record. It contains the payment amount measure and supports revenue and financial performance analysis.

Column	Type
◇ amount	double
◇ customer_id	bigint
◇ payment_date	datetime
◇ payment_id	bigint
◇ rental_id	bigint
◇ staff_id	bigint

Fact Inventory

The **fact_inventory** table stores stock level snapshots at the grain of one row per store and film combination. It contains the **stock_count** measure, which tracks the physical availability of movie copies across different locations, supporting inventory distribution analysis and store-level asset management reporting.

Column	Type
◇ film_id	bigint
◇ stock_count	bigint
◇ store_id	bigint

3.3 Dimension Tables

Customer Dimension

Column	Type
◇ active	bigint
◇ city	text
◇ country	text
◇ customer_id	bigint
◇ email	text

Film Dimension

Column	Type
◇ description	text
◇ film_id	bigint
◇ rating	text
◇ release_year	bigint
◇ rental_duration	bigint
◇ rental_rate	double
◇ title	text

Staff Dimension

Column	Type
◇ active	bigint
◇ staff_id	bigint
◇ staff_name	text
◇ store_id	bigint

4 ETL Design

The ETL (Extract, Transform, Load) process is the backbone of this project. It was implemented using **Python**, the **Pandas** library for data manipulation, and **SQLAlchemy** for database connectivity. This pipeline automates the migration from the operational sakila database to the analytical movie_rental_dw warehouse.

4.1 Extract

The extraction phase focuses on importing raw transactional data into the Python environment. We selected nine source tables to ensure all dimensions and facts could be fully

populated.
Table 5.1: Extracted Tables from OLTP

Table Name	Purpose
rental	Captures individual movie rental events
payment	Records revenue transactions
inventory	Manages stock availability
film	Contains descriptive movie attributes
staff	Contains employee data
customer	Stores customer profiles
address, city, country	Provide geographical context

4.2 Transform

Several transformation and cleaning operations were performed during the ETL process.

4.2.1 Data Cleaning

- 4.2.1.1 Metadata Pruning: We dropped last_update columns from all source tables. These columns contain operational audit timestamps that provide no value for historical business analysis.
- 4.2.1.2 Data Type Standardization: All timestamp strings (e.g., rental_date) were converted to Python datetime objects to enable temporal calculations.
- 4.2.1.3 Handling Missing Values: Missing return_date values were imputed with a placeholder date (2025-01-01). This ensures the records are preserved rather than deleted, allowing for accurate rental count metrics.

4.2.2 Data Integration

To optimize the warehouse for BI tools, we combined related tables (Denormalization):

- 4.2.2.1 Customer Dimension: Merged customer + address + city + country to enable geographical filtering without complex SQL joins.
- 4.2.2.2 Rental Fact: Merged rental + inventory to align transactions with specific film and store identifiers.

4.2.3 Calculated Measures

- 4.2.3.1 Calculated Measures: We derived duration_days from return_date minus rental_date.

Duration = Return Date – Rental Date

4.2.4 Derived Attributes

- 4.2.4.1 Derived Attributes: We concatenated first_name and last_name into a single full_name attribute for both Customers and Staff, improving readability in end-user reports.

8

4.3 Load

After transformation, the cleaned data was loaded into the movie_rental_dw database in this order:

1. Dimension Tables
2. Fact Tables

This ensures referential integrity between dimensions and facts.

4.3.1 Successfully Loaded Tables

- | | |
|------------------|----------------|
| ✓ fact_rental | ✓ dim_customer |
| ✓ fact_payment | ✓ dim_film |
| ✓ fact_inventory | ✓ dim_staff |



5 Data Quality Considerations

A Data Warehouse is only as valuable as the quality of the data it contains. To ensure a "Single Version of Truth," we implemented a multi-layered validation strategy during the ETL process. This phase ensures that the business makes decisions based on accurate, clean, and consistent information.

Quality Check	Implementation
 Missing Values	Missing return dates identified and handled properly
 Duplicate Detection	Customer and film records checked and validated
 Logical Validation	No negative rental durations verified
 Referential Integrity	Primary/foreign keys maintain consistency
 Data Consistency	Formats and naming conventions standardized

Audit Implementation:

- **Missing Value Strategy:** We identified that the operational system allowed for NULL return dates (representing active rentals). In our pipeline, we standardized these into a future-dated placeholder (2025-01-01). This prevents "Calculation Breaks" when managers attempt to run duration reports.
- **Duplicate Prevention:** Using the Pandas `.duplicated().sum()` method, we verified that each Customer and Film record in our Dimensions is unique. This ensures that metrics like "Total Revenue per Customer" are not inflated by double-counting.
- **Logical Constraint Validation:** We applied a logical check to the `duration_days` measure. By verifying that no negative values exist, we confirm that the data correctly reflects the chronological flow of time (Rent Date always occurs before Return Date).
- **Referential Integrity:** The loading order (Dimensions first, then Facts) ensures that every transaction in the Warehouse is linked to a valid business entity, maintaining the structural integrity of the Star Schema.

```
--- DATA QUALITY REPORT ---  
Missing values in Customer Names: 0  
Missing values in Rental IDs: 0  
Duplicate Customers found: 0  
Duplicate Films found: 0  
Logical Errors (Negative Days): 0  
Final Customer Dimension Shape: (599, 6)
```

6 Conclusion

- 6.1** The implementation of the Movie Rental Data Warehouse successfully transforms a complex, fragmented operational database into a high-performance analytical environment.

6.1.1 Value Delivered

- **Simplicity:** Non-technical managers can now answer complex questions (e.g., "Revenue by Country") without needing to understand the underlying 16-table Sakila structure.
- **Performance:** By denormalizing location and film data, we have reduced the SQL "Join Effort," resulting in significantly faster report generation.
- **New Insights:** The introduction of engineered metrics like `duration_days` and `stock_count` provides visibility into late-return patterns and inventory health that was previously difficult to extract from the OLTP system.